

Real-Time Drone Navigation with a Non-Stigmatizing Surface-EEG Setup and Parallel Ultra-Low-Power On-Device Computing

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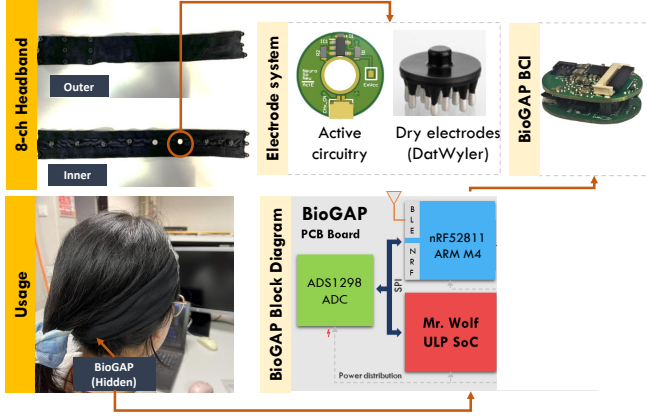


Fig. 1. Photo of headband. Electrodes (Dätwyler Schweiz AG) are integrated with an active buffering circuit, plugged into dedicated buttonholes on the headband, and connected to BioGAP [3] via cables hidden in the headband textile.

I. INTRODUCTION

We propose a novel approach to Brain-Computer Interface (BCI) systems for real-time drone navigation based on heterogeneous EEG signals, interpreted online on an energy-efficient parallel ultra-low-power (PULP) wearable platform [1]. We developed an inconspicuous EEG headband featuring 8 dry EEG channels, powered by a Machine Learning-capable PULP system to enable onboard computation, and featuring a Bluetooth low-energy wireless link [2] for direct drone-interfacing. The headband acquires, processes, and recognizes concurrently multiple user-specific EEG signals (eye blinks, eyebrows, alpha waves) thanks to a hardware-aware Machine Learning approach deployed on the platform, and translates them into drone navigation commands in real-time.

II. APPLICATION AND METHODOLOGY DESCRIPTION

Our project explores the BCI field in a new light by harnessing the EEG headband with PULP onboard computing capabilities to navigate a drone based on the user's heterogeneous EEG inputs (eye blinks, eyebrows, alpha waves). This approach enables users with more degrees of freedom to interact with technology, fostering a deeper level of engagement and extending the boundaries of potential BCI applications.

A. BioSignal acquisition system

We employed BioGAP [3], an innovative, medical-grade biosignal acquisition and processing platform. Distinguished by its compactness and modularity, the BioGAP weighs a mere 6 grams and possesses dimensions of $16 \times 21 \times 14\text{mm}^3$. It consists of two stacked printed circuit boards (PCBs). The baseboard integrates the GAP9 SoC, a ten-core ultra-low power system on a chip, along with a wireless Bluetooth Low Energy (BLE) capable SoC, a dedicated power management circuit, and an accelerometer. In contrast, the accompanying shield incorporates an analog front-end (AFE) specifically designed for ExG data acquisition. How BioGAP is integrated into the headband can be seen in Fig 1.

B. Data collection and training paradigm

We conducted a two-session data collection procedure wherein the subject performed specified EEG paradigms: double eye blink, left eye blink, right eye blink, eyebrow-raising, alpha wave generation, and baseline EEG. The data from the first session was primarily utilized for training, while the second session's data served as validation. This methodology resulted in an observed accuracy exceeding 96%. Subsequently, the final classifier was trained using data from both sessions and was integrated into the GAP9 microcontroller for real-time processing.

C. Onboard tiny Machine Learning approach and control

We extract Discrete Wavelet Transform (DWT) energy features from a 4-level DWT with the Haar mother wavelet and a temporal window length of 2s. We then calculate the energy of the detail coefficients at each level as a feature. We also calculate the FFT of the signal and look at the square root of the energy of different EEG bands. We mainly look at these features as they have been shown to work well for EEG artifact detection and seizure detection [4], [5]. We introduce a method based on gradient-boostered classifiers [6] that classify eye blinks (left eye, right eye, or both eyes), alpha waves (eyes closed), and moving eyebrows. Data for these EEG paradigms is gathered from users in real-time and used to train a Machine Learning model. Deployment and memory optimization of the trained model is done in a user-friendly way and requires only 2 minutes. A direct Bluetooth low-energy interface between the headband and the drone enables unhindered transmission

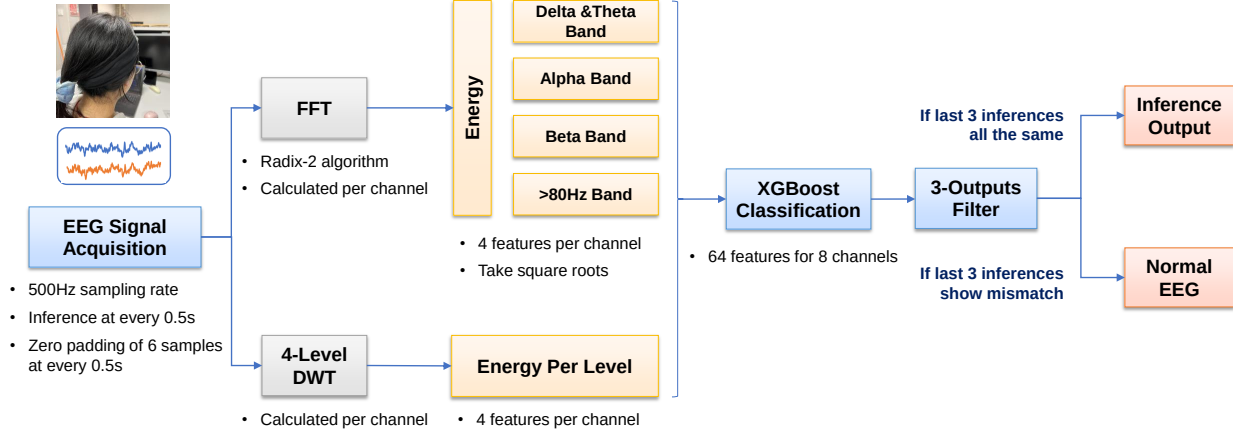


Fig. 2. EEG-based data processing pipeline: Commencing with a 500Hz sampling rate and subsequent data manipulations, leading through DWT and FFT calculations and energy summations across frequency bands, to a final XGBoost classification, culminating in drone control

of control commands in real-time, without the need to carry auxiliary devices such as heavy laptops.

D. Drone

Our BCI is coupled with the Crazyflie 2.1 quadrotor (Bitcraze), a commercial off-the-shelf (COTS) nano-drone with a weight of 27g and a diameter of 10cm, featuring an STM32F405 MCU that runs state estimation and actuation controls. The drone also incorporates a Flow deck, which provides optical flow and height measurements to increase the state estimation reliability.

III. KEY FACTS

Speed: The system performs real-time data acquisition, model training, and movement prediction onboard the GAP9 microcontroller, enabling immediate user feedback.

Accuracy: The deployed feature extraction and model showcase a very impressive 96% accuracy in the 6-class classification task.

User experience and benefits for potential users: The PULP headband ensures wearability and comfort for the user, while real-time drone navigation provides immediate and intuitive feedback [7]. The application of such a system is particularly advantageous for users with limited mobility, offering an intuitive and accessible method for device control [8]. It opens up opportunities in various fields including entertainment, healthcare, and telecommunication.

Real-Time operation: The entire pipeline, from EEG data acquisition to drone navigation, operates in real-time. The PULP platform's onboard computational capabilities enable signal classification in less than 10 ms, thereby providing an interactive and engaging user experience.

Pioneering HW/SW approach: The integration of the GAP9 microcontroller on the BioGAP platform represents a major development in BCI hardware, allowing for rapid onboard computation. In addition, the development of the

EEG classification model represents significant advancements in BCI software, enabling rapid training and subsequent deployment of the model to the GAP9.

IV. SUMMARY

The proposed PULP-headband and tiny-ML framework allow for on-device online processing at state-of-the-art energy efficiency, increasing the battery lifetime of the BCI.

Potential users: Preliminary testing will be conducted on real users to gather motor movement data. These tests will allow us to fine-tune our models, ensuring the system is robust and user-friendly.

ACKNOWLEDGMENT

This project was supported by the Swiss National Science Foundation (Project PEDESITE) This work was also supported by the ETH Future Computing Laboratory (EFCL).

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